

P1 — Analytic cohort

Week 3 · applies L2 (Analytic data) · optional and ungraded — no separate deadline, nothing to submit; M1 is assembled from these five increments Keep in your group GitHub repo: a script that rebuilds the cohort + a flow table + a short P1_notes.md.

The question this week

Can you rebuild your paper’s analytic cohort from the raw source data, and account for every person who is included or excluded? Rebuilding the sample is the first result — everything downstream depends on getting this right.

What to do

1. **Find the source files.** List every raw data file (and version/cycle) your paper’s cohort is built from, and the key you merge on.
2. **Reconstruct eligibility.** Translate the paper’s inclusion/exclusion criteria and exposure/outcome definitions into code. Build the exposure and link the outcome.
3. **Make a flow table.** Report N after each exclusion step, ending at your analytic N.
4. **Compare to the paper and explain the gap.** State your N vs the paper’s. Trace any difference to a specific decision. **Do not tune your code to hit the paper’s number.**

Deliverable

- Reproducible cohort-building script (runs start-to-finish on a fresh machine).
- A flow table (CSV or figure) → becomes your M1 **cohort flow diagram**.
- P1_notes.md: your one-sentence study question, and a numbered list of every eligibility decision with the reasoning where the paper was ambiguous.

Worked example — MASLD/NHANES demonstration

- Six NHANES cycles + linked mortality merged on SEQN; FLI computed; four BMI/WHtR groups formed. **Not a target — a comparison.** The paper reports $N \approx 6,300$, groups 386 / 4,541 / 769 / 604 and 585 deaths; our reproduction’s **full analytic file** lands on **N = 6,371**, groups **391 / 4,585 / 787 / 608** and **586 deaths**, and its **locked domain** — the rows carrying a valid fasting weight — is **6,048**. Getting 6,371 rather than the paper’s number is not a failure: it is the finding, and tracing *why* the two differ is the skill being taught. An unexplained exact match is worth less than a discrepancy you can account for.
- **Trap 1 — one sentence, three cohorts.** “Significant alcohol (>2 drinks/day men, >1 women)” coded three defensible ways gives **N = 4,308 / 5,944 / 6,371**. The paper’s numbers match *no effective exclusion*. Which did the authors use? You cannot tell from the paper — so you **document the ambiguity**, you don’t force it.
- **Trap 2 — a table label that disagrees with the Methods.** The Table 1 headers say “BMI < 25”, but the non-obese groups have BMI medians of 27.6 and 28.8. The operative cut is **30** (from the Methods). Coding to 30 reproduces the groups; coding to 25 does not.

What good looks like

- **Exceptional:** full step-by-step funnel; the residual N gap is *traced*, and any part that cannot be explained is reported as unresolved (here, the alcohol codings were tested and none of them accounts for the gap); ambiguities documented rather than tuned away.
- **Adequate:** cohort rebuilt and runs; N close; most exclusions justified.
- **Weak:** “N matches ” with no funnel, or code that hard-codes the target N.

Be ready to defend (M3)

“Show me one person who is in your cohort but not the paper’s — and tell me why.”

AI co-pilot note

A co-pilot will happily write code that filters until N equals the paper’s number. That is the exact failure this increment tests. You must be able to show the funnel and defend each cut.

Lab → project handoff

You are applying **L2 (Analytic data)** — multi-source download, merge, cleaning, eligibility, and exposure/outcome construction (EpiMethods `accessingE3` and `accessingE2`). Same moves, your paper’s data.